

School of Engineering

Cochin University of Science & Technology

B.Tech Degree 2nd semester Second Internal Examination, July 2024

23-200-0202 A Engineering Physics (2023 scheme)

(CE A, CE B/ME B, SF)

Time: 2 hours

Max.Marks: 30

Course Outcomes:

1. Interpret modern devices and technologies based on lasers and optical fibres
2. Explain the basic principles of crystal physics
3. Summarise the characteristic and applications of superconducting materials, nanomaterials and smart materials
4. Illustrate the theory of semiconductors and magnetic materials
5. Understand the principle, concept, working and applications of relevant technologies and comparison of results with theoretical calculations.

PART A (Answer all questions) (5x2=10)

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|--|---|----|---|---|
| 1.a. Explain 'quantum confinement effect ' | 2 | L2 | 3 | 1 |
| b. Discuss any two important applications of superconductors | 2 | L3 | 3 | 3 |
| c. What is Meissner effect | 2 | L2 | 3 | 2 |
| d. Discuss the terms Magnetic Susceptibility and magnetic permeability | 2 | L3 | 4 | 3 |
| e. Differentiate between extrinsic and intrinsic semiconductors | 2 | L1 | 4 | 1 |

PART B (2x10=20)

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|---|---|----|---|---|
| 11 a. Define Superconductivity. Explain BCS Theory of superconductivity | 7 | L2 | 3 | 1 |
| b. Calculate the critical current that flow through a long thin superconducting wire of aluminium of diameter 10^{-2} m. The critical magnetic field for aluminium is 7.9×10^3 amp. per metre. | 3 | L3 | 3 | 2 |

OR

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|--|---|----|---|---|
| 111.a. Discuss the optical, electrical and magnetic properties of metallic glasses | 7 | L2 | 3 | 2 |
| b. Discuss any three important applications of shape memory alloys | 3 | L3 | 3 | 3 |
| IV a. Describe the properties of para, dia and ferro magnetic materials. | 7 | L2 | 4 | 2 |
| b. A solenoid has a core of a material with relative permeability 400. Windings of the solenoid are insulated from the core and carry a current of 2A. if the number of turns is 1000per metre. Calculate H, M and B | 3 | L3 | 4 | 3 |

OR

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|--|---|----|---|---|
| V a. Discuss the effect of temperature on semiconductors | 7 | L2 | 4 | 1 |
| b. C (atomic number 6), Si (atomic number 14) and Ge (atomic number 32) have the same lattice structure. Why is C insulator while Si and Ge are intrinsic semiconductors | 3 | L3 | 4 | 3 |

$J \propto \eta I$
 $J = \eta \times I$

$B = \mu_0 (H + J)$